4-Channel IR Sensor - 16-State Table

Physical sensor order: P1 P2 P3 P4, left to right across the sensor bar. 1 = black line detected; 0 = white / no black line.

P1 P2 P3 P4	Class	Interpretation	Line offset	Inner speed	Operator / steering action
P0000	AMBIGUOUS	Blind band or line lost	-	0.40	Use previous reading to resolve
P0001	DRIFT	Far right, outer sensor only	+3.2 cm	0.13	Steer right strongly
P0010	DRIFT	Drifting right, slight	+1.0 cm	0.73	Steer right slightly
P0011	JUNCTION	Right pair; needs >2.8 cm black	+1.8 cm	0.40	Junction / curve evidence
P0100	DRIFT	Drifting left, slight	-1.0 cm	0.73	Steer left slightly
P0101	NOISE	Split dark regions	-	1.00	Hold previous command
P0110	ON LINE	Centred	0.0 cm	1.00	Drive straight
P0111	JUNCTION	Branch or curve on the right	+1.6 cm	0.40	Junction / curve evidence
P1000	DRIFT	Far left, outer sensor only	-3.2 cm	0.13	Steer left strongly
P1001	NOISE	Outer pair only	-	1.00	Hold previous command
P1010	NOISE	Split dark regions	-	1.00	Hold previous command
P1011	NOISE	P2 dropped out	-	1.00	Hold previous command
P1100	JUNCTION	Left pair; needs >2.8 cm black	-1.8 cm	0.40	Junction / curve evidence
P1101	NOISE	P3 dropped out	-	1.00	Hold previous command
P1110	JUNCTION	Branch or curve on the left	-1.6 cm	0.40	Junction / curve evidence
P1111	JUNCTION	Symmetric crossbar	0.0 cm	1.00	Junction / crossbar evidence

Important: P0000 is context-dependent. After P0010 or P0100 it is the 0.8 cm blind band and steering continues toward the previous offset. After P0001 or P1000 it indicates genuine line loss. P0110 is the centred straight-line state. Non-contiguous patterns are noise and must not steer on their own.

Source: src/carbot/ir_geometry.py, STATE_TABLE. Print at 100% / Actual Size on A4; do not use Fit to Page if exact physical spacing is needed.